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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	Duplicate Record Detection System	
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Abstract

- The Duplicate Record Detection System is a data science project designed to identify and remove duplicate or highly similar records from datasets. Duplicate records can reduce data quality, produce inaccurate analysis, and affect the performance of machine learning models. The proposed system uses Python and data science techniques to automatically detect duplicate entries based on attributes such as names, email addresses, phone numbers, addresses, and other relevant fields.

- The system uses Python libraries such as Pandas and NumPy for data preprocessing and manipulation, along with techniques such as data normalization, string matching, similarity measurement, and duplicate detection. Exact duplicates can be identified using Pandas functions, while fuzzy or near-duplicate records can be detected by comparing the similarity between text fields. The system preprocesses the dataset by handling missing values, removing unnecessary spaces, standardizing text formats, and converting data into a consistent structure before detecting duplicates.

- The main objective of this project is to develop an efficient and automated solution that improves *data accuracy, consistency, and reliability. The system can be applied to customer databases, student records, healthcare datasets, transaction records, and other real-world datasets where duplicate information is common. The final output provides a cleaned dataset with duplicate records identified or removed, helping organizations maintain high-quality data for further analysis and decision-making.

- **Technologies Used**
  Python, Pandas, NumPy, data preprocessing, string matching, and similarity-based duplicate detection

Signature of the Guide HOD